Project Proposal

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Dataset

Source: https://data-explorer.oecd.org/vis?lc=en&tm=ICT&pg=0&snb=69&df%5Bds%5D=dsDisseminateFinalDMZ&df%5Bid%5D=DSD_ICT_HH_IND%40DF_IND&df%5Bag%5D=OECD.STI.DEP&df%5Bvs%5D=1.1&dq=.A........&pd=,&to%5BTIME_PERIOD%5D=false

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Description: The dataset contains information on ICT (information and communication technologies) usage among individuals across multiple indicators, such as age, sex, education level, income group, and purpose of ICT use.

It is important to note that a pre-filtered version of the dataset will be used in the following analysis to reduce storage size (the full dataset is over 500MB).

The following filters will be applied:

- Time period: 2021–2025
- Reference areas: Germany, United States, France

This pre-filtered dataset contains 42 columns and 19010 rows. Each row represents a unique combination of multiple indicators, such as age and education level, and an observed value representing a part of the population or a count of people. It can be found in `data/raw` and is called: `ICT_data_raw.csv`.

Research Questions

1. Is education level associated with the proportion of the population that uses the internet daily or almost every day within different age groups?

   For the done analysis take a look in: 4.1_analysis.qmd in code/

   Needed columns: AGE, EDUCATION_LEVEL, OBS_VALUE, MEASURE

   This question is explanatory but not causal. A possible correlation could be driven by other factors, such as income differences across education levels resulting in differing access to some technologies.

2. How has the proportion of the population using the internet daily or almost every day changed between 2021 and 2025 across different countries, and does this trend differ by employment status or income group?

Context and target audience

The results of the analysis could be of interest to businesses, as they may help to understand which categories of audiences are reachable through the internet and how these groups interact with the internet.

A clean website presenting the results using professional language and visualizations would be appropriate.

Initial data analysis

```
Rows: 19,010
Columns: 42
$ STRUCTURE                  <chr> "DATAFLOW", "DATAFLOW", "DATAFLOW", "DATAFL~
$ STRUCTURE_ID               <chr> "OECD.STI.DEP:DSD_ICT_HH_IND@DF_IND(1.1)", ~
$ STRUCTURE_NAME             <chr> "ICT Access and Usage by Individuals", "ICT~
$ ACTION                     <chr> "I", "I", "I", "I", "I", "I", "I", "I", "I"~
$ REF_AREA                   <chr> "FRA", "FRA", "FRA", "FRA", "FRA", "FRA", "~
$ `Reference area`           <chr> "France", "France", "France", "France", "Fr~
$ FREQ                       <chr> "A", "A", "A", "A", "A", "A", "A", "A", "A"~
$ `Frequency of observation` <chr> "Annual", "Annual", "Annual", "Annual", "An~
$ MEASURE                    <chr> "D1A_I", "F4G_I", "G2I_I", "F2A_I", "G2G_I"~
$ Measure                    <chr> "Individuals using the internet for e-
maili~
$ UNIT_MEASURE               <chr> "PT_POP", "PT_POP", "PT_POP", "PT_POP", "PT~
$ `Unit of measure`          <chr> "Percentage of population", "Percentage of ~
$ GEO_AREA                   <chr> "_Z", "_Z", "_Z", "_Z", "_Z", "_Z", "_Z", "~
$ `Geographical area`        <chr> "Not applicable", "Not applicable", "Not ap~
$ AGE                        <chr> "Y16T74", "Y16T74", "Y16T74", "Y16T74", "Y1~
```

```
$ Age                          <chr> "From 16 to 74 years", "From 16 to 74 years~
$ SEX                          <chr> "_T", "_T", "_T", "_T", "_T", "_T", "_T", "~
$ Sex                          <chr> "Total", "Total", "Total", "Total", "Total"~
$ EDUCATION_LEVEL              <chr> "HI", "HI", "HI", "HI", "HI", "HI", "HI", "~
$ `Education level`            <chr> "High level of educational attainment", "Hi~
$ INCOME_GROUP                 <chr> "_T", "_T", "_T", "_T", "_T", "_T", "_T", "~
$ `Income group`               <chr> "Total", "Total", "Total", "Total", "Total"~
$ EMP_STATUS                   <chr> "_T", "_T", "_T", "_T", "_T", "_T", "_T", "~
$ `Employment status`          <chr> "Total", "Total", "Total", "Total", "Total"~
$ TIME_PERIOD                  <dbl> 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2~
$ `Time period`                <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ OBS_VALUE                    <dbl> 95.8430, 1.8295, 10.3323, 76.2974, 18.3431,~
$ `Observation value`          <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ OBS_STATUS                   <chr> "A", "A", "A", "A", "A", "A", "A", "A", "A"~
$ `Observation status`         <chr> "Normal value", "Normal value", "Normal val~
$ OBS_STATUS_2                 <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ `Observation status 2`       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ OBS_STATUS_3                 <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ `Observation status 3`       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ UNIT_MULT                    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ `Unit multiplier`            <chr> "Units", "Units", "Units", "Units", "Units"~
$ TIME_HORIZON_USE             <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ `Time horizon`               <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ DECIMALS                     <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2~
$ Decimals                     <chr> "Two", "Two", "Two", "Two", "Two", "Two", "~
$ BREAKDOWN_V7_HH              <chr> "I_HI", "I_HI", "I_HI", "I_HI", "I_HI", "I_~
$ `V7 Breakdowns households` <chr> "Individuals aged 16-74 with a high level o~
```

```r
data_raw %>%
  dplyr::select(AGE, EDUCATION_LEVEL, TIME_PERIOD, OBS_VALUE, Measure, REF_AREA) %>%
  summary()
```

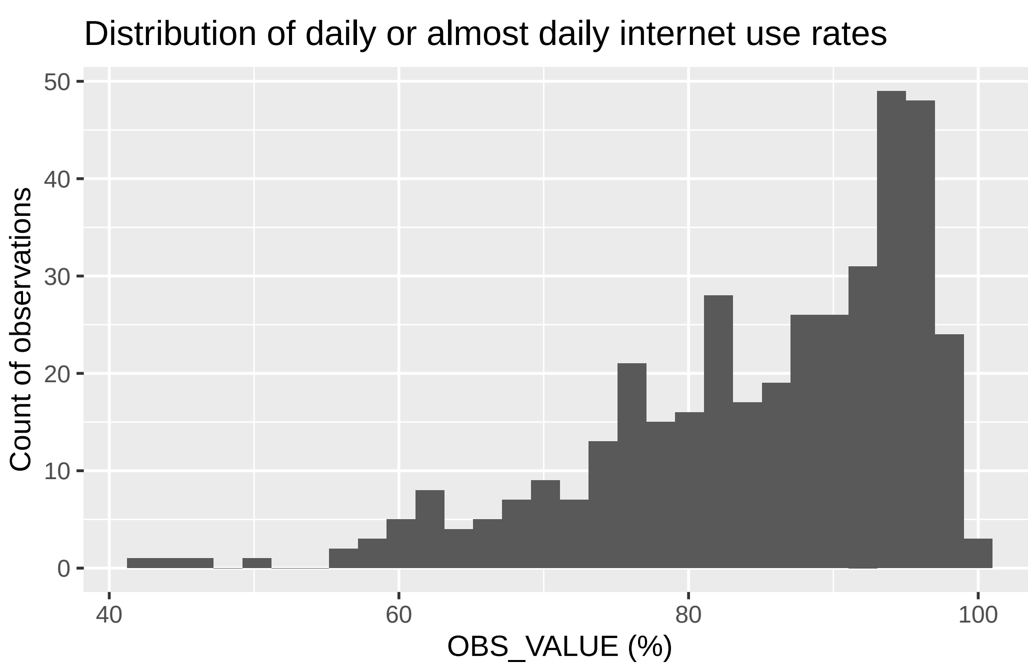
```
     AGE            EDUCATION_LEVEL     TIME_PERIOD     OBS_VALUE        
 Length:19010       Length:19010       Min.   :2021   Min.   :        0  
 Class :character   Class :character   1st Qu.:2021   1st Qu.:       10  
 Mode  :character   Mode  :character   Median :2023   Median :       30  
                                       Mean   :2023   Mean   :   180110  
                                       3rd Qu.:2024   3rd Qu.:       60  
                                       Max.   :2025   Max.   :243380968  
   Measure            REF_AREA        
 Length:19010       Length:19010      
 Class :character   Class :character  
```

```
 Mode  :character   Mode  :character
```

Plot 1 — Distribution of the outcome variable:

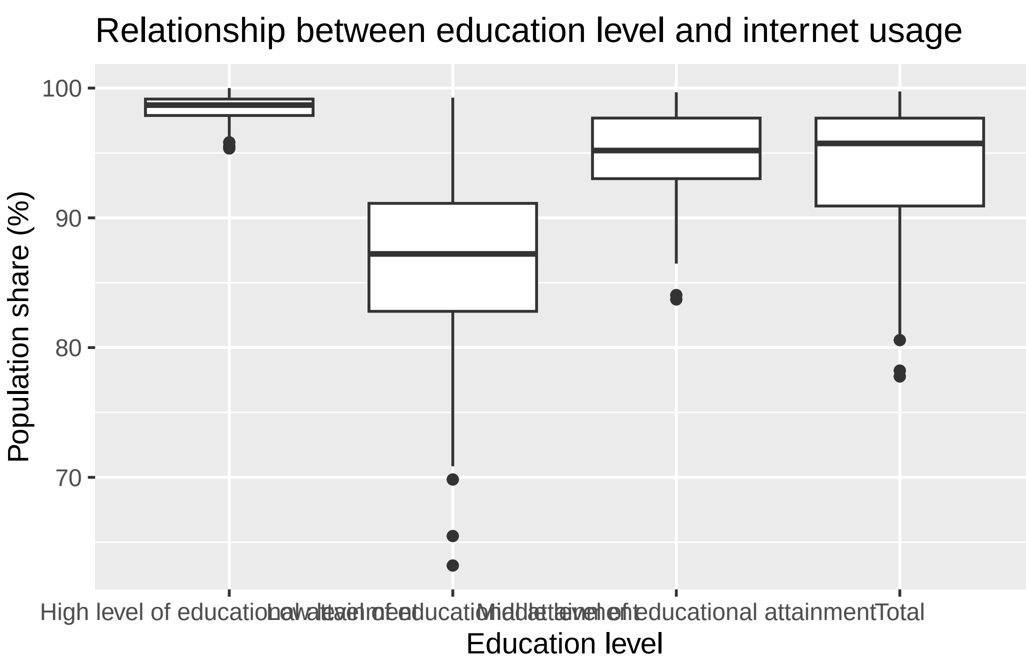
```
data_raw %>%
  filter(MEASURE == "C5B1_I") %>%
  filter(UNIT_MEASURE == "PT_POP") %>%
  ggplot(aes(x = OBS_VALUE)) +
  geom_histogram() +
  labs(
    title = "Distribution of daily or almost daily internet use rates",
    x = "OBS_VALUE (%)",
    y = "Count of observations"
  )
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Plot 2 — Relationship or trend relevant to a research question:

```r
data_raw %>%
  filter(MEASURE == "C5A_I") %>%
  ggplot(aes(x = `Education level`, y = OBS_VALUE)) +
  geom_boxplot() +
  labs(
    title = "Relationship between education level and internet usage",
    x = "Education level",
    y = "Population share (%)",
  )
```



Data quality issues:

- Some indicators rule each other out. This may be due to missing observations for certain combinations of indicators. For example, education level and employment status rule each other out.
- As a result, questions such as "How does education level influence employment status…" are not answerable with this dataset due to the absence of observations containing both indicators.
- There are also many useless columns. Some contain the same information in different formats, while others are entirely filled with NA values.

- The documentation is lackluster. The only documented indicator is MEASURE/ Measure. The remaining indicators are fortunately mostly self-explanatory or not needed for the analysis, but it is still important to note the lack of documentation.

Analysis Plan

- *Which variables will you use?*
  - Research Question 1: AGE, EDUCATION_LEVEL, OBS_VALUE, MEASURE
  - Research Question 2: OBS_VALUE, MEASURE, INCOME_GROUP, EMP_STATUS, TIME_PERIOD, REF_AREA
- *What analytical approach or method seems appropriate, and why?*
  - Question 1: We will regress `OBS_VALUE` (the share of the population using the internet daily or almost every day) on `EDUCATION_LEVEL`, `AGE`, and their interaction using linear regression. We will report a coefficient table with confidence intervals and a boxplot with `AGE` as facet. This addresses RQ1 because it quantifies the association between education level and daily internet use and assesses whether this association differs across age groups.
  - We will first create line plots showing how the proportion of the population using the internet daily or almost every day changed between 2021 and 2025 across different countries. We will then use faceted line plots to compare these trends across employment status or income groups. This exploratory analysis will help assess whether changes in internet use over time differ between employment or income groups.
- *What cleaning or transformation steps do you anticipate needing?*

  Categorical variables need to be declared as such during import. Columns that contain the same information in different formats can be removed. Columns that are not needed for the analysis can be dropped.
- *What are the main uncertainties or risks (e.g. not enough data, confounding variables, missing values)?*

  The biggest risk is the absence of certain indicator combinations. The research questions will need to be planned around this limitation. This dataset will not be able to answer every possible question based on the available indicators.

Workflow organisation

- *Code style:* We will follow the coding style taught in StatProg1 and StatProg2, including the use of the tidyverse style with piping (%>%), the here package for file paths, and other conventions. We will enforce this using a code_style.md file, which documents used packages and coding style conventions.
- *Packages:* tidyverse, janitor, here, …
- *Git workflow:* There will always be at least two branches to avoid conflicts within the group. This, together with a clear separation of tasks, should help to prevent merge conflicts. We will also work on separate .qmd files, which will later be combined into a final document using the include command for additional modularity.
- *`.gitignore`:*

  .Rhistory .RData .Rproj.user/

  /.quarto/ _site/

  .DS_Store Thumbs.db

  **/*.quarto_ipynb

  …